



"Your Partner in Innovative Refrigeration and Advanced Testing Solutions."

Walk-in Test Chamber

Sri Easwari Scientific Solution Pvt Ltd.
Chennai.

 +91 94444 27748

+91 94444 59430

 info@sess.co.in

 www.sess.co.in

SESS Walk in chambers is designed for simulating, monitoring & control of environmental conditions like temperature & humidity. All these chambers are double walled & modular in construction, easy to assemble at site. They are available in standard sizes & can also be tailor made to suit specific requirements. Forced air circulation with the help of fan blower maintains uniform conditions. Use of intelligent control systems & dedicated SESS Climatic Viewer' software complying to 21 CFR Part 11 requirements greatly enhances the performance. These chambers can be provided with standby refrigeration & humidity systems (optional) for uninterrupted functioning.

Construction

- Double walled prefabricated modular panels with PUF insulation, easy to assemble at site.
- Standard model with inner of Stainless Steel (304 Grade) and outer Powder coated steel sheet.
- GMP model with inner (316 Grade) and outer of Stainless Steel (304 Grade)
- CFC free Refrigeration system with split type condensing unit located outside lab helps to emit hot air beyond working area.
- Heating by long lasting SS tubular heaters.
- Humidity by steam injection. Boiler and water reservoir made Of SS with heater and low water level cut Off.
- Forced Air Circulation by heavy duty motor • SS racks and trays.

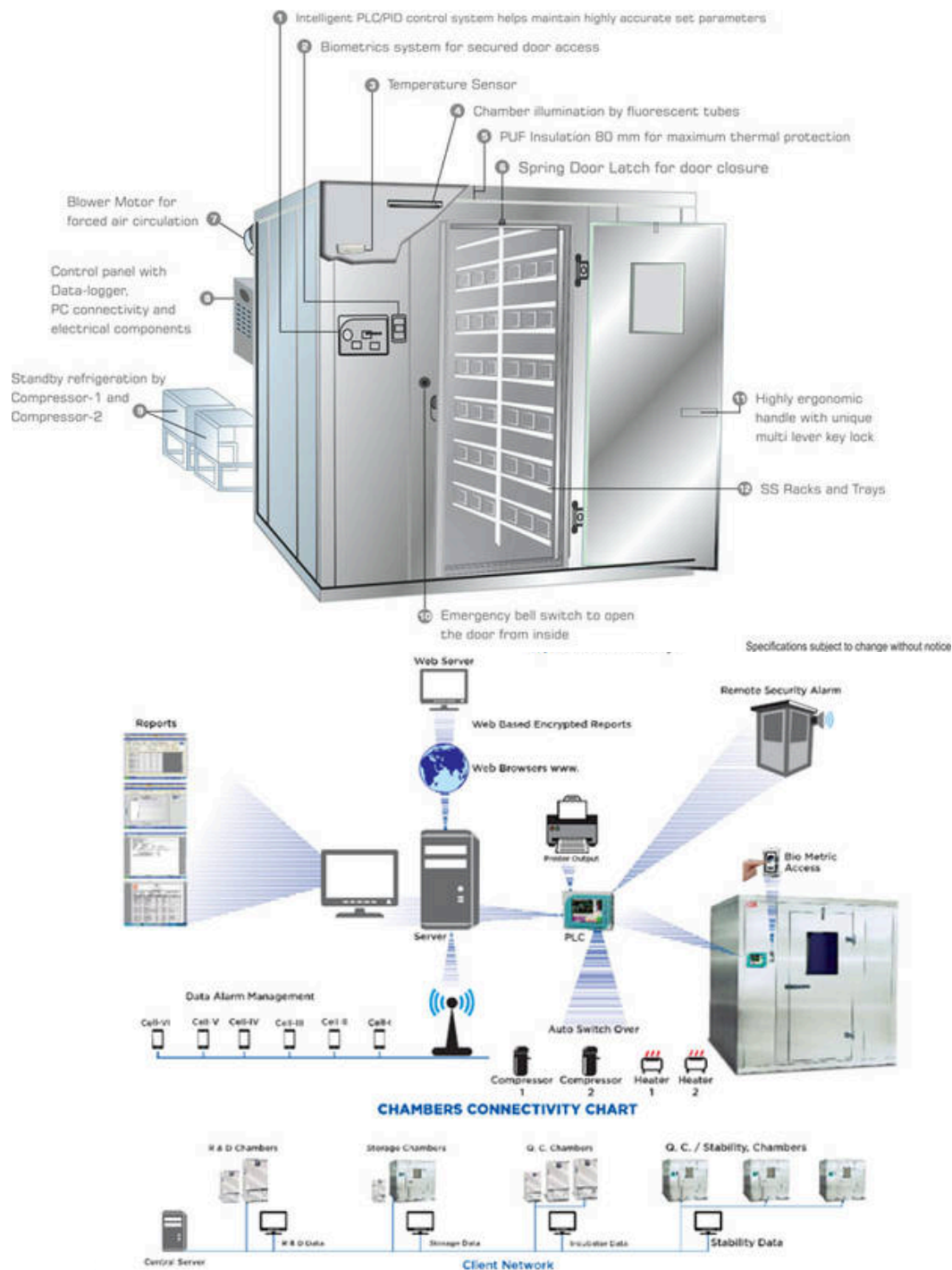


Salient Features

- Designed as per ICH guidelines to meet, WHO and USFDA requirement.
- Fulfills storage conditions of 25°C-60% RH, 40°C-75% RH, 30°C-65% RH 25°C-40% RH, 30°C-35% RH
- Temperature Range: 20°C to 60°C, Accuracy: $\pm 0.2^{\circ}\text{C}$, Uniformity: $\pm 1^{\circ}\text{C}$ (Stability Chambers).
- Humidity Range: 40% RH to 95% RH, Accuracy: $\pm 2\%$ RH, Uniformity: $\pm 3\%$ RH (Stability Chamber).
- Temperature Range: 2°C to 8°C, Accuracy: $\pm 0.2^{\circ}\text{C}$, Uniformity: $\pm 1^{\circ}\text{C}$ (Cold Rooms)
- Microprocessor based auto tuning PID controller provided for precise control of temperature and humidity.
- Auto Change over to standby systems through PLC (Optional)
- Auto switch over to standby sensor in case of controlling sensor failure through PLC (Optional).
- Capacitance type Humidity sensor direct display of RH and temperature (Stability Chambers)
- Audio visual alarm for temperature, humidity variations and utility failures.
- Output for remote alarm in lab and at security gate.

- Safety includes shut off of humidity and air heaters in case of temperature overshoots or undershoots beyond specified limits with alarms.
- Facility to open the door from inside in case door locked with emergency bell having switch inside chamber.
- Low water level alarm and power cut off of boiler tank heater.
- Overload cut off relay for compressor
- Time delay for compressor switch on.
- PC communication via Ethernet/RS-485 and software complying to 21CFR part 11 USFDA guidelines (Optional).
- Data stored in PC server with LAN connectivity and password protection (Optional with software)

- PLC with touch screen HMI fully protected with password, instead of PID controller for precise control of temperature and humidity (Optional).
- Standby refrigeration system available for uninterrupted functioning (Optional).
- Standby humidity system available for uninterrupted functioning (Optional).
- GSM mobile alert in case of temperature and humidity deviations beyond specified Limits (Optional with software).
- Magnetic door lock system with biometrics for door access and software to monitor door open log (Optional).
- Validation protocol with IQ, OQ, PQ Documentation.



LCD DISPLAY

- 4 Inch attractive LCD display
- Intelligent Controller helps maintain temperature in case of sensor failure
- Battery Backup for Temperature Controller
- Auto tuning of controller
- Self-Diagnosis for errors
- Histogram format of 24 hours temperature recording
- Power failure and resumption recorded with date and time
- Settable High / low alarm points
- Memory for storage of upto 1000 records
- Centronic interface to record Temperature, Date, & Time by attaching Dot Matrix printer with adjustable print interval
- Alarm on equipment / output for remote alarm

TOUCH SCREEN DISPLAY

- 7 inch big colour touch screen
- PLC (Programable logic controller)
- Memory of 10,000 records
- Centronic interface to record Temperature, Date & Time by attaching Dot Matrix printer with adjustable print interval
- Ethernet port
- Battery backup in case of power failure for display, alarms & off-line data logging (optional)
- Programmable cyclic timer for controlling illumination conditions
- PC Communication through SESS soft

SESS 'DATA SOFT'

- Multiple level alpha numerical password with password ageing
- Secured audit trail report
- Alarm report with mention of alarm condition
- Mean Kinetic Temperature able to be calculated for any days & alarm deviation report
- Reports in graphical and tabular form
- Data Stored in PC server with RS-485 connectivity and can be monitored on PC with password authentication
- Multi chamber data acquisition on single software
- Communication through RS 485 / TCP IP
- GSM Mobile Alert for six Numbers

OPTIONAL ACCESSORIES

- Data logger for 8 / 16 Point temperature & Humidity recording with sensors printer interface and software
- Software as per 21 CFR Part 11 requirements
- PLC Based Control System with HMI Touch Screen
- Biometrics Magnetic Door Access for Secured Door opening with software to maintain log
- Validation (mapping) consist of one cycle for temperature at any one set point for 24 hours on empty and loaded condition, Standby Humidity System
- Standby Refrigeration System,
- Standby humidity System
- GSM alert through SIM card

Technical Data					
WALK-IN STABILITY CHAMBERS	ETC - 50 P	ETC - 80 P	ETC - 100 P	ETC - 120 P	ETC - 150 P
WALK-IN COLD ROOM	WCR-50	WCR-80	WCR-100	WCR-120	WCR-150
Internal Volume (Liters)	5000	8000	10000	12500	15000
Internal Dimensions W*D*H (mm)	16	24	32	48	48
External Dimensions W*D*H (mm)	1250*2000*2000	2000*2000*2000	2000*2000*2500	2000*2500*2400	2000*2500*2400
Internal Body Material	1410*2360*2160	2160*2360*2160	2160*2360*2660	2160*2860*2560	2660*2860*2560
External Body Material	Stainless Steel 304 grade (Standard Models), Stainless Steel 316 Grade (GMP Models)				
Insulation	Powder Coated CRCA Steel (Standard Models), Stainless Steel 304 Grade (GMP Models)				
Illumination	80mm minimum , CFC free polyurethane foam				
Control & Display	Fluorescent Tubes				
Power Failure Alarm	Microprocessor based PID control system with large 4' LCD Display				
Door Open Alarm	Audio Visual Alarm				
	Audio Alarm in case door open for over one minute				

Supply: 220 - 240 Volts, 50Hz Single Phase

	WALK-IN STABILITY CHAMBERS	WALK-IN COLD ROOM
Sensor	Capacitance type direct reading	PT-100
Temperature Range & Accuracy	20°C to 60°C, +- 1°C	2°C to 8°C, +- 1°C
Humidity Range (RH) & Accuracy	40% to 90%, +-3% RH	xx
Audio Visual Variation Alarm	Set Humidity = 3% RH, Set Temperature= 2°C	Set Temperature = 2°C



THANK YOU!

| Presented By
SESS TECHNICAL TEAM

Sri Easwari Scientific Solution Pvt Ltd.

No. 2/298, ANE Garden, Perumal Kovil Street, Paraniputhur Post,
Iyyappanthangal, Chennai - 600 122.

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